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STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Tajikistan: Joint Bank-Fund Debt Sustainability Analysis

Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Unsustainable under current policies*
Application of judgment	*No*

This joint World Bank/ IMF Debt Sustainability Analysis (DSA) has been prepared, in the context of the 2019 Article IV Consultation. This DSA is based on a new framework for low-income countries[1]. Results indicate that Tajikistan's risk of debt distress is high under the baseline scenario for both external and public debt, similar to the results of the 2017 DSA.[2] Tajikistan's current debt-carrying capacity is assessed as medium. However, Tajikistan's debt is on an unsustainable path under current policies. Further, it is vulnerable, especially to export shocks, currency depreciation, and contingent fiscal liabilities. Medium-term fiscal consolidation, avoiding non-concessional external borrowing, diversifying exports, and containing contingent liabilities from SOEs would help reduce vulnerabilities.

[1] See IMF, 2018, Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-Income Countries.

[2] Tajikistan is assessed to have a medium debt-carrying capacity. CI rating for Tajikistan is 2.832, which is based on the April 2019 WEO and the World Bank's 2018 CPIA.

COVERAGE AND BACKGROUND ON PUBLIC DEBT

A. Background on Debt and Debt Coverage

1. In recent years, external and financial sector vulnerabilities have contributed to an increase in debt. Tajikistan's external public- and publicly-guaranteed (PPG) debt rose from 24 percent of GDP in 2014 to near 39 percent of GDP at end-2018 mainly as a consequence of a sizable depreciation of the somoni as well as the issuance of a $500 million sovereign bond in 2017. Domestic PPG debt also increased from 3½ percent of GDP at end-2014 to 9 percent of GDP at end-2018, owing to a 6 percent of GDP recapitalization of banks in December 2016. As a result, the total PPG debt increased from 27½ percent of GDP in 2014 to 47.9 percent of GDP in 2018.

2. External debt on concessional terms made up the bulk of the total of PPG debt in 2018. External PPG debt accounted for about 82 percent of total PPG debt. Over 80 percent of external PPG debt was owed to multilateral and bilateral creditors on concessional terms. The single largest creditor was China, which held over 40 percent of the total PPG external debt. Non-concessional debt consists only the Eurobond of USD 500 million issued in September 2017, with a maturity of 10 years, carried an interest rate of 7.125 percent.

3. The NBT is the main creditor and holder of largely non-marketable domestic government debt. Most of the government securities held by the NBT were issued at significantly below market terms, with some interest rates as low as 0.99 percent. Since 2016, the government has been accumulating interest and principal arrears to the NBT. In 2019, the arrears on domestic government securities issued for the NBT recapitalization were cleared after the NBT extended new credit to the government at a 2 percent interest rate with a one-year maturity. However, the government continues to run arrears against the NBT on bonds issued to recapitalize commercial banks during the 2015-16 shocks.

4. This DSA covers the central government, central bank, and government-guaranteed external and domestic debt. As of September 2019, debt coverage includes duly consolidated overall external and domestic debt and guarantees of the Central Government (CG), including extrabudgetary funds, and the social security fund. As debt recording and monitoring capacity is weak, this DSA does not include in its baseline: i) non-guaranteed liabilities of state-owned enterprises (SOEs), including liabilities associated with the modernization of an aluminum plant and the construction of a gas pipeline[3], (ii) contingent

[3] The Ministry of Finance does not record non-guaranteed debt of SOEs. TALCO, a loss-making state-owned aluminum company, was recently allowed to borrow without a government guarantee. Subsequently, TALCO has signed a MOU to borrow $545 million from China to modernize its plant. The finalization of contract and associated disbursement are expected to in 2020. Separately, Tajiktransgaz has borrowed $300 million from Chinese entities for the construction of the Tajikistan section of the Turkmenistan-China gas pipeline.

liabilities/fiscal costs associated with potential liquidation of two large and troubled financial institutions or iii) demand or guarantees triggered from any existing PPP agreements.[4]

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

BACKGROUND ON MACRO FORECASTS

5. The assumptions in the baseline scenario are consistent with macroeconomic framework presented in the Staff Report. The main assumptions are:

- ***External.*** The current account deficit is expected to remain sizable over the medium term with high fiscal deficits and limited exchange rate flexibility to facilitate external adjustment. Remittances are expected to remain sluggish owing to the global environment. Electricity exports are expected to rise owing to Roghun over the medium term. International reserves decline over the projection horizon.

- ***Interest rates***. Effective average interest rates on external debt are projected to rise gradually over the medium-term as concessional financing is likely to be constrained and the authorities would increasingly rely on non-concessional external borrowing. Interest rates on domestic public debt, some of which are highly negative in real terms at present, are expected to remain below marketable rates.

- ***Fiscal***. The fiscal deficit is expected to remain high over the medium-term, in line with authorities' plans for spending on Roghun and other large infrastructure projects.

- ***Growth.*** While growth will be supported in the near-term by government spending, a weak global environment, continued large fiscal and external deficits, and uneven structural reforms are expected to weigh on growth over the medium term. Inflation is expected to remain moderate.

6. Macroeconomic assumptions under the current baseline scenario project a higher fiscal deficit than under the 2017 DSA. While real GDP growth estimates for 2017-18 were larger than projected in 2017, nominal GDP estimates are broadly similar. A key difference is that the projected fiscal deficit has risen as projections now incorporate the full extent of government spending on Roghun. Recapitalization costs for the two formerly systemic banks were included in the fiscal projections in 2017-18, but are not

[4] State and local governments are not allowed to borrow without a government guarantee.

expected at this time as liquidation is considered to be more appropriate. The external position was projected to remain weak in 2017 as well as in the current DSA. International reserves (in months of imports) are higher in the current DSA owing to domestic purchases of gold, but are expected to decline gradually over the medium-term.

Tajikistan: Baseline DSA Assumptions, 2017-24

(In percent of GDP, unless otherwise indicated)

	2017	2018	2019	2020	2021	2022	2023	2024
Real GDP growth, percent								
Current DSA	7.1	7.3	5.5	4.8	4.5	4.5	4.0	4.0
Previous DSA	4.5	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Overall fiscal balance (incl. PIP)								
Current DSA	-6.0	-2.8	-3.8	-4.3	-4.3	-4.3	-4.3	-4.3
Previous DSA	-6.5	-5.4	-1.9	-1.8	-1.7	-1.7	-2.0	-2.2
Current account balance								
Current DSA	2.2	-5.0	-4.5	-5.2	-5.3	-5.3	-5.4	-5.5
Previous DSA	-6.3	-6.2	-5.6	-5.1	-4.6	-4.3	-4.3	-4.2

Sources: National authorities and IMF staff estimates

7. The baseline scenario assumes that fiscal financing needs will be met from external concessional sources in the near-term, but non-concessional financing will rise over time. In 2019, the remainder of Eurobond proceeds issued in 2017 are available for disbursement. During 2020-22, fiscal financing needs are expected to be met mostly from concessional loans and grants, provided by international finance institutions, export credit agencies, and traditional bilateral partners. Budget support loans and grants are projected to remain low. There is uncertainty on the terms of future borrowing. For the purposes of this DSA, staff assume concessional borrowing after 2022 in line with the levels of recent years. Residual financing needs are expected to be met from non- concessional borrowing. It should be noted that if the authorities contract non-concessional borrowing, a waiver from the World Bank may be needed under the Non-concessional Borrowing Policy clause. Staff projections assume no new domestic financing at market determined rates, in line with the recent experience.

8. The realism tools largely suggest that staff forecasts are realistic. Under the baseline scenario, debt accumulation over the projection horizon is much smaller than in recent years. An important reason for the difference is that the contribution of exchange rate depreciation to external and public debt accumulation is much lower than in recent years, as no large external shocks (as in 2015-16) are assumed under the baseline. Staff projections of growth, while lower than estimates from recent years, are in line with weaker external demand going forward.[5] It is noteworthy that the contribution of primary deficits to future debt accumulation is expected to be similar to the past few years. Although debt accumulation over the next

[5] There are weaknesses in national accounts statistics.

5 years is slower than the historic 5 years under the baseline, both external and public debt are on an increasing trajectory throughout the projection horizon.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

9. Tajikistan is assessed to have a medium debt-carrying capacity. Tajikistan's debt carrying capacity is currently assessed as medium based on the CI Index from the April 2019 WEO and the World Bank's 2018 CPIA. The 2017 DSA assessed Tajikistan's debt carrying capacity as weak. The upgrade from weak to medium debt-carrying capacity is mainly driven by a shift from CPIA to a composite indicator, which also incorporates other factors such as the import coverage of reserves and remittances.

Debt Carrying Capacity	Medium

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.83	Medium 2.90	Medium 2.97

10. Stress tests for PPPs' agreements, potential size of a rescue of the financial sector, and a commodity price shock are set at default levels. Stress test for PPPs' demand and guarantees is set at a default 1.73 percent of GDP. Stress test for the banking sector is set at default 5 percent of GDP. Default tailored tests for commodity prices are also applied since non-fuel commodity exports constitute an important part of Tajikistan's exports.

11. Tailored contingent liability stress test is designed to incorporate contingent liabilities from potential non-guaranteed debt of SOEs. The debt coverage for Tajikistan excludes non-government guaranteed debt of non-financial public corporations (NFPC) under the baseline given uncertainties on the nature of the debt and lack of full financial information on SOEs. To illustrate the effects of contingent liabilities associated with large SOE debt that might have significant implications for debt sustainability, the size of shock is set at 10 percent of GDP. The shock reflects: (i) 6.5 percent of GDP based on available information on Barki Tojik arrears, which could be transferred onto the government's balance sheet; and (ii) 3.5 percent of GDP based on the loan agreements of Tajiktransgaz. Planned borrowings related to the modernization of the TALCO aluminum plant could pose additional contingent liability risks (6½ percent of GDP). However, it is not clear at this stage if the contract to finance the aluminum plant will materialize and what the contract would look like, including whether it would be considered debt, so it is not included in this DSA.

1	The country's coverage of public debt	The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	10	Great uncertainty about the true size of liabilities and weak financial position and performance
4	PPP	35 percent of PPP stock	1.73	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)	16.7		

DEBT SUSTAINABILITY

B. External Debt Sustainability Analysis

12. Under the baseline scenario, three external debt indicators breach respective thresholds (Figure 1). Baseline breaches involve both stock/solvency (PV debt-to-GDP ratio, PV debt-to-export ratio) and flow/liquidity (debt-service-to exports ratio) indicators. Breaches are significant and protracted and do not fall below the thresholds throughout projection horizon. Both solvency indicators rise steadily throughout the projection horizon. The flow indicator breaches the threshold during Eurobond principal repayment period and stays above the threshold. Fiscal consolidation, aligning exchange rate with fundamentals, prudent borrowing policies, and better debt management could help address these challenges and reduce the risk of debt distress.

13. Under the stress scenarios, three external debt indicators breach respective thresholds. All breaches are significant and point to debt vulnerabilities. Shocks to exports and contingent liabilities are the most extreme and impactful. Under a shock to exports, the PV of debt-to-exports reaches about 435 percent by the end of horizon. The contingent liability shock also causes a significant deterioration in external debt sustainability. The trajectory of the PV of external public debt-to-GDP ratio moves upwards by close to 12 percentage points from the baseline. This suggests the need for the government to improve debt recording and management practices (especially for SOEs) and rebuild fiscal buffers to address the rising contingent liabilities. The market-financing risks are moderate in the medium-term because the repayment of Eurobond issuance is not due until 2024. Additional commercial borrowing to refinance Eurobond repayments and meet gross financing needs is not expected to become due before 2029, alleviating market-financing risk for the projection horizon. The market financing risk could become significant in the outer years, especially as the necessity to meet gross financing needs through increasing borrowing may widen sovereign spreads.

C. Public Sector Debt Sustainability Analysis

14. Under the baseline scenario, the public debt burden indicator (PV total debt-to-GDP) ratio remains below the 55 percent benchmark. However, the ratio rises throughout the projection horizon.

15. The standardized sensitivity analysis shows that a shock to commodity prices breaches the threshold and leads to the highest public debt figures in 2029. The shock causes a 12 percent deterioration in debt ratios by 2029. The contingent liabilities shock is also sizable and causes an 11 percent deterioration in debt ratios. This highlights the need for strengthened oversight of SOE sector and streamlined borrowing policies at a time when the government is already financing a large infrastructure project.

AUTHORITIES' VIEWS

16. The authorities concurred with staff that debt vulnerabilities need to be better managed but considered the DSA assessment as unduly negative. They considered staff's growth projections as conservative and expected annual real GDP growth to be higher than the staff's baseline projection in line with recent experience owing to their industrialization strategy. They agreed with staff that non-concessional borrowing would weaken debt sustainability and would examine options to attract concessional financing. Staff emphasized the need for fiscal consolidation to ensure debt sustainability.

17. There was broad agreement that better oversight on the SOEs is needed to contain contingent liabilities. However, the authorities maintained that debt contracted by SOEs, including TALCO and Tajiktransgaz do not pose fiscal risks. In their view, these companies would be financially viable to repay debt.

CONCLUSION AND RISK RATING

18. The debt sustainability analysis under the new LIC DSF framework suggests that Tajikistan's risk of external and overall public debt distress is high. These results are similar to the 2017 DSA findings.

19. Tajikistan's risk of external debt distress remains high. Three external debt-burden indicators breach respective thresholds under the baseline, pointing to potential solvency and liquidity issues. External debt is most vulnerable to contingent liabilities and exports shocks. Baseline scenario and standardized stress tests indicate the importance of containing contingent liabilities and broadening the export base.

20. Overall risk of public debt distress is high under the baseline due to breaches in external debt indicators. A shock to commodity prices has the largest impact on public debt sustainability.

21. Under current policies Tajikistan's public debt is on an unsustainable path. Three external debt indicators breach respective thresholds under the baseline. Two stock indicators are on a continuously increasing trajectory while one flow indicator breaches and remains above the threshold.

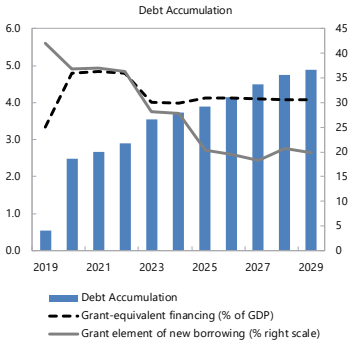
22. Measures should be taken to reduce debt vulnerabilities. Embarking on fiscal consolidation, relying on concessional loans and avoiding non-concessional borrowing would help contain debt servicing costs and the debt burden. Diversifying exports and containing contingent liabilities will reduce vulnerabilities of public debt to shocks. Improving debt management practices, upgrading the debt recording and reporting practices would further help to contain debt vulnerabilities.

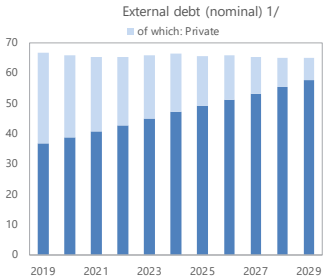
Table 1. Tajikistan: External Debt Sustainability Framework, Baseline Scenario, 2016-2039

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**67.4**	**76.9**	**70.5**	**66.7**	**65.8**	**65.2**	**65.2**	**65.9**	**66.5**	**64.9**	**80.0**	**56.1**	**65.6**
of which: public and publicly guaranteed (PPG)	**32.1**	**40.6**	**38.7**	**36.9**	**38.8**	**40.7**	**42.8**	**45.0**	**47.2**	**57.6**	**76.4**	**32.3**	**47.1**
Change in external debt	5.4	9.5	-6.4	-3.8	-0.8	-0.6	0.0	0.7	0.6	-0.2	1.2		
Identified net debt-creating flows	**9.3**	**-4.9**	**-2.1**	**-2.1**	**-0.2**	**0.5**	**0.8**	**1.4**	**1.5**	**0.8**	**1.3**	**2.2**	**0.5**
Non-interest current account deficit	**2.8**	**-3.9**	**4.3**	**2.6**	**3.1**	**3.2**	**3.3**	**3.4**	**3.4**	**3.1**	**2.3**	**5.8**	**3.2**
Deficit in balance of goods and services	29.2	22.9	28.0	25.6	26.0	25.7	25.5	25.4	25.4	25.1	25.9	**38.4**	**25.4**
Exports	12.9	15.7	14.8	15.4	15.8	16.3	16.6	16.8	16.9	17.2	18.5		
Imports	42.1	38.7	42.8	41.0	41.8	42.0	42.1	42.2	42.2	42.2	44.4		
Net current transfers (negative = inflow)	-8.2	-8.2	-6.6	-6.3	-6.2	-6.1	-6.1	-6.1	-6.1	-6.1	-6.4	**-9.4**	**-6.1**
of which: official	-1.7	-1.4	-0.9	0.0	-0.4	-0.3	-0.5	-0.5	-0.5	-0.5	-0.5		
Other current account flows (negative = net inflow)	-18.1	-18.6	-17.0	-16.7	-16.7	-16.5	-16.1	-15.9	-15.9	-15.9	-17.3	**-23.2**	**-16.1**
Net FDI (negative = inflow)	**-3.0**	**-0.9**	**-3.3**	**-3.0**	**-2.4**	**-2.1**	**-1.8**	**-1.6**	**-1.6**	**-2.2**	**-1.4**	**-3.0**	**-2.2**
Endogenous debt dynamics 2/	**9.5**	**-0.2**	**-3.1**	**-1.7**	**-0.9**	**-0.7**	**-0.7**	**-0.4**	**-0.4**	**-0.1**	**0.5**		
Contribution from nominal interest rate	1.4	1.6	0.8	1.9	2.1	2.1	2.0	2.1	2.1	2.4	3.4		
Contribution from real GDP growth	-4.8	-4.7	-5.3	-3.6	-3.0	-2.8	-2.8	-2.5	-2.5	-2.5	-3.0		
Contribution from price and exchange rate changes	12.9	2.9	1.5	...	...	...	...	...	...	...	...		
Residual 3/	**-3.9**	**14.4**	**-4.3**	**-1.7**	**-0.6**	**-1.1**	**-0.8**	**-0.7**	**-0.8**	**-1.0**	**-0.1**	**2.1**	**-1.0**
of which: exceptional financing	0.0	0.0	0.0	0.0	-0.4	-0.3	-0.5	-0.5	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**29.6**	**27.9**	**28.9**	**29.7**	**30.8**	**32.6**	**34.4**	**44.9**	**65.9**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**199.6**	**181.1**	**182.9**	**182.4**	**185.7**	**193.6**	**204.0**	**261.3**	**357.0**		
PPG debt service-to-exports ratio	**12.3**	**12.7**	**16.3**	**15.8**	**16.6**	**15.4**	**14.3**	**12.2**	**12.0**	**17.0**	**34.1**		
PPG debt service-to-revenue ratio	**5.8**	**7.4**	**9.2**	**10.0**	**10.7**	**10.2**	**9.7**	**8.4**	**8.2**	**11.9**	**25.7**		
Gross external financing need (Million of U.S. dollars)	1351.6	832.2	1584.5	1317.0	1682.5	1736.4	2233.8	2689.5	3132.3	3604.8	4681.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.9	7.1	7.3	5.5	4.7	4.5	4.5	4.0	4.0	4.0	4.0	**6.7**	**4.3**
GDP deflator in US dollar terms (change in percent)	-17.2	-4.1	-1.9	2.5	0.2	1.8	1.4	1.3	1.4	1.7	1.7	**-2.1**	**1.6**
Effective interest rate (percent) 4/	2.0	2.5	1.0	2.9	3.3	3.4	3.3	3.3	3.4	3.8	4.6	**1.8**	**3.4**
Growth of exports of G&S (US dollar terms, in percent)	9.1	25.1	-0.8	12.2	7.8	9.5	7.8	7.0	5.7	6.7	7.3	**8.7**	**7.4**
Growth of imports of G&S (US dollar terms, in percent)	-12.0	-5.6	16.5	3.6	7.0	6.8	6.0	5.7	5.6	5.8	7.5	**-0.1**	**5.8**
Grant element of new public sector borrowing (in percent)	...	...	...	42.0	36.8	37.0	36.3	28.2	27.8	19.9	11.0	**...**	**27.9**
Government revenues (excluding grants, in percent of GDP)	27.4	27.0	26.2	24.5	24.5	24.5	24.5	24.5	24.5	24.5	24.5	**24.6**	**24.5**
Aid flows (in Million of US dollars) 5/	171.5	187.3	216.6	343.1	445.9	486.3	493.6	382.3	403.4	540.3	767.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.3	4.8	4.8	4.8	4.0	4.0	4.1	3.7	**...**	**4.2**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	68.1	56.6	57.1	56.4	50.3	49.8	42.4	30.8	**...**	**49.8**
Nominal GDP (Million of US dollars)	6,953	7,144	7,520	8,132	8,539	9,081	9,619	10,130	10,683	14,160	24,828		
Nominal dollar GDP growth	-11.5	2.8	5.3	8.1	5.0	6.4	5.9	5.3	5.5	5.8	5.8	**4.5**	**5.9**
Memorandum items:													
PV of external debt 7/	...	...	61.4	57.6	56.0	54.2	53.2	53.5	53.8	52.2	69.6		
In percent of exports	...	...	413.6	374.2	354.0	333.0	321.1	317.5	318.6	303.5	376.8		
Total external debt service-to-exports ratio	151.6	104.1	135.4	107.6	120.3	110.5	131.0	147.1	162.8	143.1	97.6		
PV of PPG external debt (in Million of US dollars)			2227.8	2267.9	2470.4	2697.8	2960.5	3302.2	3678.9	6357.7	16371.2		
(PVt-PVt-1)/GDPt-1 (in percent)				0.5	2.5	2.7	2.9	3.6	3.7	4.9	5.4		
Non-interest current account deficit that stabilizes debt ratio	-2.6	-13.3	10.7	6.5	3.9	3.8	3.3	2.7	2.8	3.3	1.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Eα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, E=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

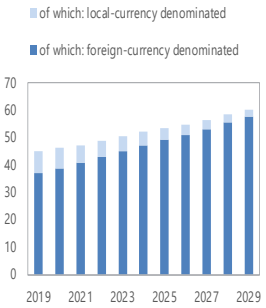
Table 2. Tajikistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016-2039

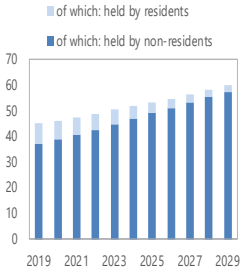
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**42.1**	**50.4**	**47.9**	**45.1**	**46.1**	**47.3**	**48.8**	**50.5**	**52.0**	**60.3**	**78.1**	**37.2**	**52.1**
of which: external debt	**32.1**	**40.6**	**38.7**	**36.9**	**38.8**	**40.7**	**42.8**	**45.0**	**47.2**	**57.6**	**76.4**	**32.3**	**47.1**
Change in public sector debt	7.4	8.3	-2.5	-2.8	1.0	1.2	1.4	1.7	1.6	1.9	1.5		
Identified debt-creating flows	**14.7**	**3.6**	**-1.0**	**0.4**	**1.1**	**0.9**	**0.8**	**0.9**	**0.7**	**1.1**	**0.5**	**32.7**	**0.8**
Primary deficit	**8.3**	**5.5**	**1.7**	**2.7**	**3.2**	**3.3**	**3.5**	**3.5**	**3.4**	**3.9**	**3.4**	**2.4**	**3.4**
Revenue and grants	29.9	29.7	29.1	26.7	27.2	27.2	27.1	27.0	27.0	27.2	27.2	**27.1**	**27.1**
of which: grants	2.5	2.6	2.9	2.2	2.7	2.7	2.7	2.5	2.4	2.7	2.7		
Primary (noninterest) expenditure	38.3	35.2	30.8	29.4	30.3	30.6	30.6	30.4	30.4	31.1	30.6	**29.5**	**30.6**
Automatic debt dynamics	**0.1**	**-0.2**	**-2.2**	**-1.5**	**-2.1**	**-2.5**	**-2.7**	**-2.6**	**-2.7**	**-2.8**	**-2.9**		
Contribution from interest rate/growth differential	-2.0	-3.1	-4.2	-1.5	-2.1	-2.5	-2.7	-2.6	-2.7	-2.8	-2.9		
of which: contribution from average real interest rate	0.3	-0.3	-0.8	1.0	-0.1	-0.5	-0.6	-0.7	-0.7	-0.6	0.1		
of which: contribution from real GDP growth	-2.2	-2.8	-3.4	-2.5	-2.0	-2.0	-2.0	-1.9	-1.9	-2.2	-2.9		
Contribution from real exchange rate depreciation	2.1	2.9	2.0	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**6.2**	**-1.7**	**-0.4**	**-0.8**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**31.6**	**-0.1**
Privatization receipts (negative)	0.0	-1.0	-0.7	-0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	6.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.1	-0.7	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-7.3**	**4.7**	**-1.5**	**-3.2**	**0.0**	**0.3**	**0.6**	**0.8**	**0.8**	**0.8**	**1.0**	**-30.7**	**0.3**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**39.7**	**37.1**	**36.9**	**36.9**	**37.4**	**38.7**	**40.0**	**48.5**	**69.0**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**136.5**	**139.0**	**135.8**	**135.6**	**137.7**	**143.5**	**148.2**	**178.1**	**253.4**		
Debt service-to-revenue and grants ratio 3/	**6.2**	**9.2**	**10.8**	**11.2**	**12.3**	**11.9**	**11.5**	**10.2**	**9.8**	**10.8**	**24.6**		
Gross financing need 4/	16.4	6.6	4.4	4.9	6.5	6.6	6.6	6.2	6.0	6.8	10.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.9	7.1	7.3	5.5	4.7	4.5	4.5	4.0	4.0	4.0	4.0	**6.7**	**4.3**
Average nominal interest rate on external debt (in percent)	0.0	1.5	1.3	1.8	2.6	2.5	2.1	2.1	2.1	2.2	3.2	**0.3**	**2.2**
Average real interest rate on domestic debt (in percent)	2.5	0.0	0.1	-4.2	-2.9	-5.2	-5.2	-5.0	-5.2	-5.7	-5.2	**-2.4**	**-5.1**
Real exchange rate depreciation (in percent, + indicates depreciation)	7.3	9.7	5.3	...	...	...	...	...	...	...	...	**2.9**	**...**
Inflation rate (GDP deflator, in percent)	5.3	4.7	5.0	6.5	6.3	6.2	5.7	5.6	5.7	6.1	6.1	**7.5**	**6.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	30.2	-1.5	-6.2	0.7	8.2	5.3	4.7	3.4	3.7	4.1	3.7	**8.5**	**4.4**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.9	-2.8	4.1	5.5	2.1	2.2	2.0	1.8	1.8	2.0	1.9	**0.8**	**2.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

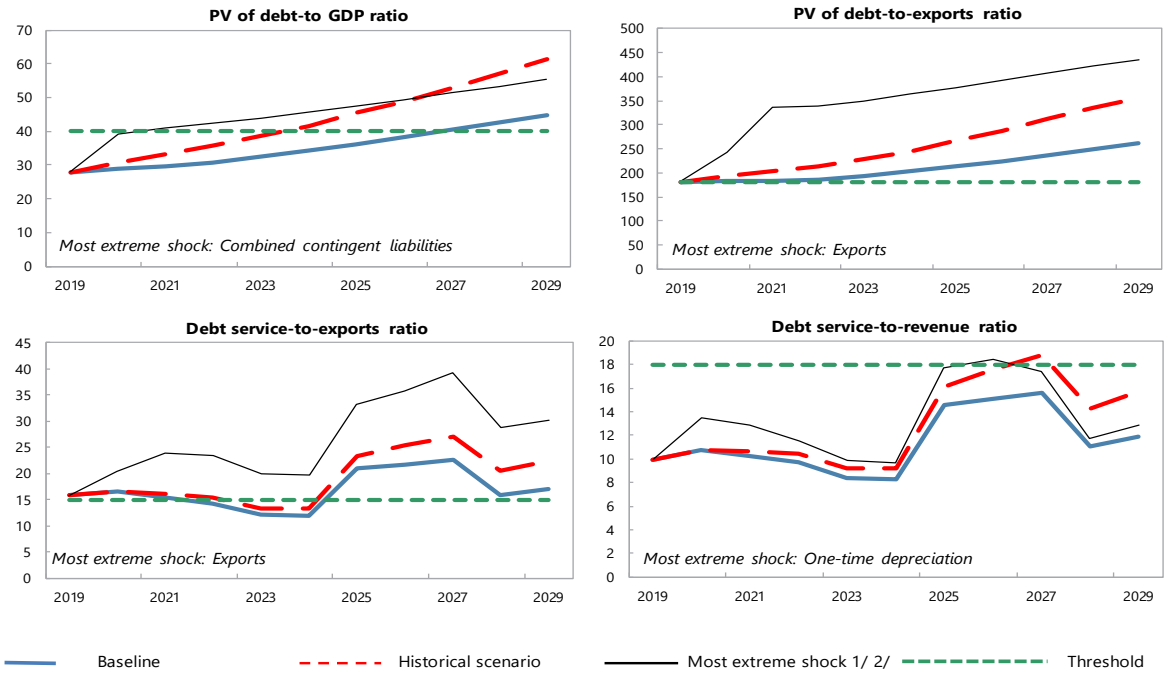
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2019–29



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	No	No
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	2.2%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	5	5

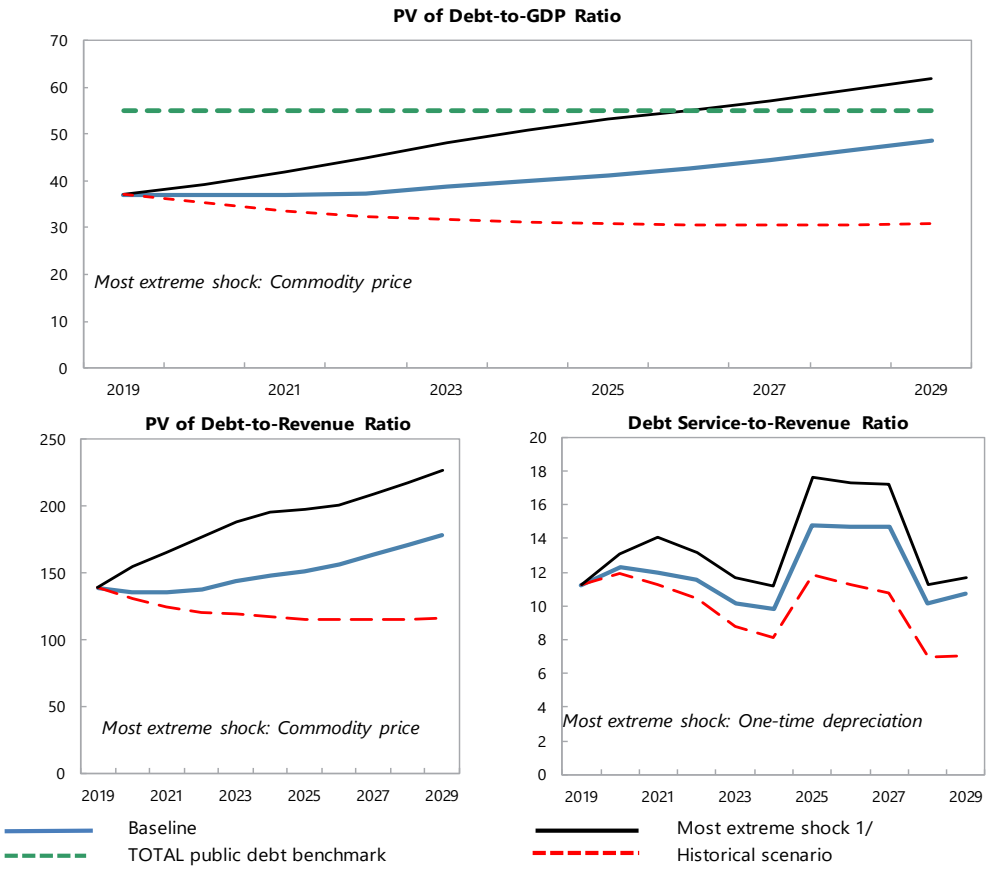
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Tajikistan: Indicators of Public Debt Under Alternative Scenarios, 2019–29



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	94%	94%
Domestic medium and long-term	0%	0%
Domestic short-term	6%	6%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	2.2%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-1.1%	-1.1%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019-2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	28	29	30	31	33	34	36	38	**40**	**43**	**45**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	28	31	33	36	38	**41**	**45**	**49**	**53**	**57**	**61**
B. Bound Tests											
B1. Real GDP growth	28	29	31	32	34	36	37	39	**42**	**44**	**46**
B2. Primary balance	28	31	36	37	38	**40**	**42**	**44**	**46**	**48**	**50**
B3. Exports	28	32	37	38	40	**42**	**43**	**45**	**47**	**49**	**51**
B4. Other flows 3/	28	31	35	36	37	39	**41**	**43**	**44**	**46**	**48**
B5. Depreciation	28	37	31	32	35	37	39	**42**	**45**	**49**	**52**
B6. Combination of B1-B5	28	33	34	35	37	38	**40**	**42**	**44**	**46**	**49**
C. Tailored Tests											
C1. Combined contingent liabilities	28	39	**41**	**42**	**44**	**46**	**47**	**49**	**51**	**53**	**56**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	28	31	34	35	36	37	38	39	40	**41**	**42**
C4. Market Financing	28	29	30	31	33	34	36	38	**40**	**43**	**45**
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	**181**	**183**	**182**	**186**	**194**	**204**	**213**	**225**	**238**	**250**	**261**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	**181**	**194**	**204**	**215**	**228**	**245**	**267**	**287**	**311**	**334**	**356**
B. Bound Tests											
B1. Real GDP growth	**181**	**183**	**182**	**186**	**194**	**204**	**213**	**225**	**238**	**250**	**261**
B2. Primary balance	**181**	**195**	**218**	**222**	**229**	**238**	**246**	**258**	**269**	**280**	**290**
B3. Exports	**181**	**243**	**336**	**339**	**349**	**365**	**377**	**393**	**408**	**423**	**436**
B4. Other flows 3/	**181**	**198**	**212**	**214**	**222**	**232**	**240**	**251**	**261**	**271**	**281**
B5. Depreciation	**181**	**183**	150	154	163	174	**183**	**196**	**211**	**226**	**239**
B6. Combination of B1-B5	**181**	**220**	**197**	**244**	**253**	**265**	**276**	**289**	**303**	**317**	**330**
C. Tailored Tests											
C1. Combined contingent liabilities	**181**	**248**	**253**	**255**	**261**	**271**	**279**	**290**	**302**	**313**	**323**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**181**	**230**	**236**	**231**	**229**	**230**	**229**	**234**	**240**	**246**	**252**
C4. Market Financing	**181**	**183**	**182**	**186**	**194**	**204**	**213**	**225**	**238**	**250**	**261**
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	16	17	15	14	12	12	21	22	23	16	17
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	16	17	16	15	13	13	23	25	27	21	22
B. Bound Tests											
B1. Real GDP growth	16	17	15	14	12	12	21	22	23	16	17
B2. Primary balance	16	17	16	16	14	14	23	24	26	19	20
B3. Exports	16	20	24	23	20	20	33	36	39	29	30
B4. Other flows 3/	16	17	16	15	13	13	22	23	25	18	19
B5. Depreciation	16	17	15	13	11	11	20	21	20	13	15
B6. Combination of B1-B5	16	18	19	18	15	15	26	28	29	21	22
C. Tailored Tests											
C1. Combined contingent liabilities	16	17	17	16	14	14	23	23	24	17	19
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	16	20	18	17	14	14	23	24	26	18	18
C4. Market Financing	16	17	15	14	12	12	21	22	23	16	17
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	10	11	10	10	8	8	15	15	16	11	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	10	11	11	10	9	9	16	18	**19**	14	16
B. Bound Tests											
B1. Real GDP growth	10	11	11	10	9	9	15	16	16	11	12
B2. Primary balance	10	11	11	11	10	9	16	17	18	13	14
B3. Exports	10	11	11	11	9	9	16	17	**18**	14	14
B4. Other flows 3/	10	11	10	10	9	9	15	16	17	13	13
B5. Depreciation	10	14	13	12	10	10	18	**18**	17	12	13
B6. Combination of B1-B5	10	11	11	10	9	9	15	17	17	12	13
C. Tailored Tests											
C1. Combined contingent liabilities	10	11	11	11	10	9	16	16	17	12	13
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	10	12	12	11	10	9	16	16	17	12	13
C4. Market Financing	10	11	10	10	8	8	15	15	16	11	12
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Tajikistan: Sensitivity Analysis for Key Indicators of Public Debt, 2019-2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	37	37	37	37	39	40	41	43	44	46	49
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	37	35	34	32	32	31	31	31	31	31	31
B. Bound Tests											
B1. Real GDP growth	37	38	39	40	42	44	46	48	50	53	**56**
B2. Primary balance	37	39	43	44	45	46	47	49	50	52	54
B3. Exports	37	39	43	44	45	46	47	49	50	51	53
B4. Other flows 3/	37	39	42	42	44	45	46	47	49	50	52
B5. Depreciation	37	42	39	38	37	37	37	36	37	38	39
B6. Combination of B1-B5	37	37	38	38	39	40	40	41	43	44	46
C. Tailored Tests											
C1. Combined contingent liabilities	37	49	49	49	51	52	53	54	**56**	**58**	**59**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	37	39	42	45	48	51	53	**55**	**57**	**60**	**62**
C4. Market Financing	37										
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	139	136	136	138	143	148	151	157	163	171	178
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	139	130	125	120	119	118	116	115	115	116	117
B. Bound Tests											
B1. Real GDP growth	139	139	143	147	156	163	168	176	184	194	203
B2. Primary balance	139	144	160	161	167	171	174	179	184	191	198
B3. Exports	139	144	159	161	167	171	174	179	183	189	194
B4. Other flows 3/	139	145	154	156	161	166	169	173	178	184	191
B5. Depreciation	139	155	146	140	140	138	135	135	136	139	143
B6. Combination of B1-B5	139	136	141	141	145	148	149	153	157	162	168
C. Tailored Tests											
C1. Combined contingent liabilities	139	182	180	181	187	192	194	199	205	211	218
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	139	155	165	177	188	195	198	201	209	217	226
C4. Market Financing	139										
Debt Service-to-Revenue Ratio											
Baseline	11	12	12	12	10	10	15	15	15	10	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	11	12	11	10	9	8	12	11	11	7	7
B. Bound Tests											
B1. Real GDP growth	11	12	12	12	11	11	16	16	16	11	12
B2. Primary balance	11	12	13	13	11	11	16	16	17	12	13
B3. Exports	11	12	12	12	11	10	15	16	17	12	13
B4. Other flows 3/	11	12	12	12	11	10	15	16	16	12	12
B5. Depreciation	11	13	14	13	12	11	18	17	17	11	12
B6. Combination of B1-B5	11	12	12	12	10	10	15	15	15	10	11
C. Tailored Tests											
C1. Combined contingent liabilities	11	12	17	13	11	11	16	16	16	11	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	11	13	13	14	13	12	17	17	18	13	14
C4. Market Financing											

Sources: Country authorities; and staff estimates and projections.

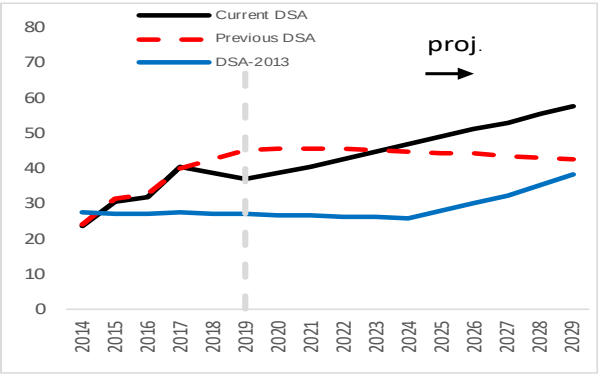
1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

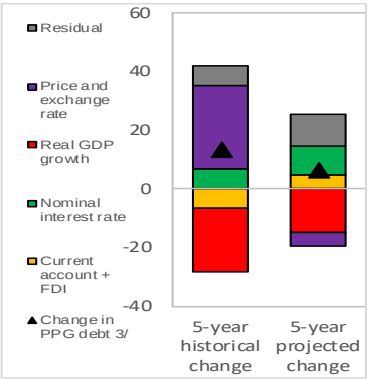
3/ Includes official and private transfers and FDI.

Figure 3. Tajikistan: Drivers of Debt Dynamics – Baseline Scenario External Debt

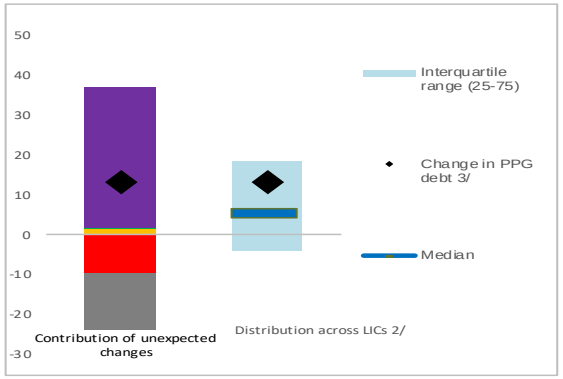
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

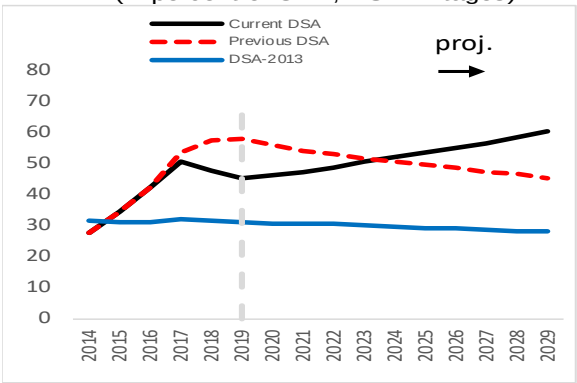


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

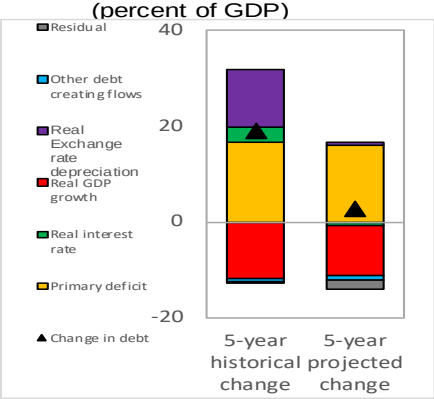


Public debt

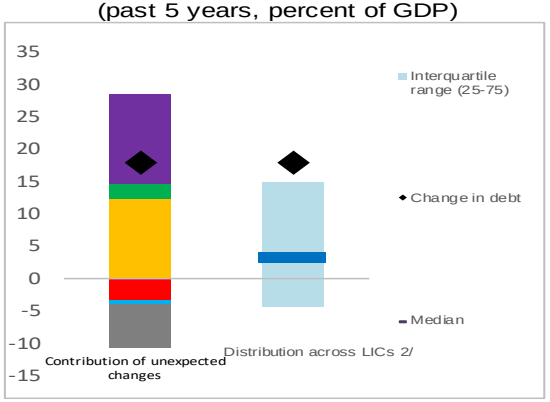
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



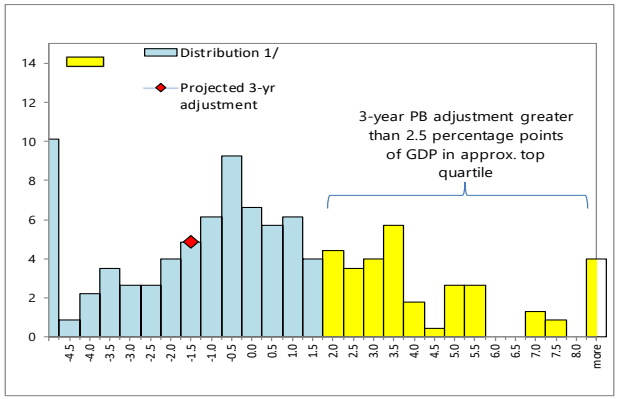
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

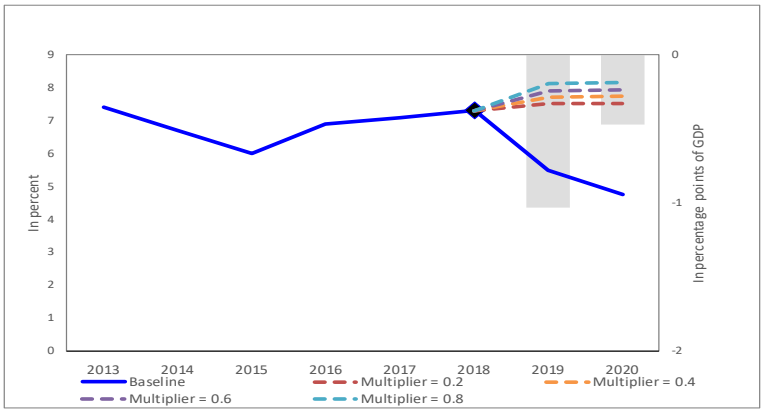
Figure 4. Tajikistan: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



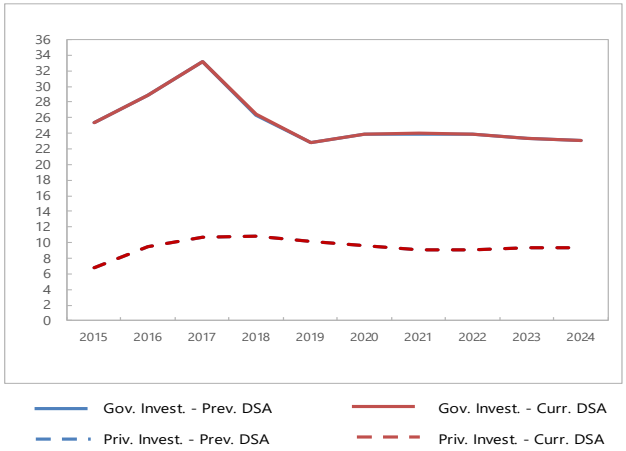
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)

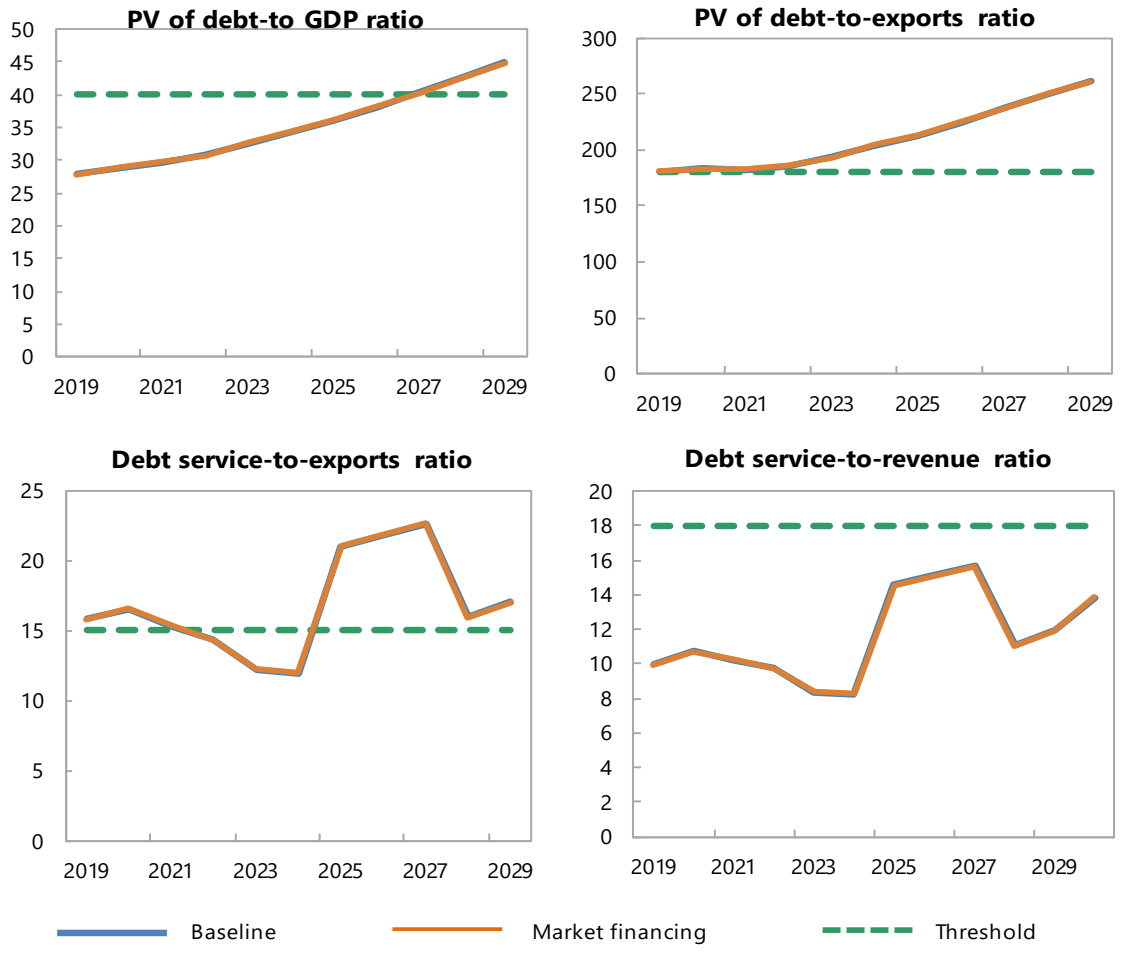


Contribution to Real GDP growth
(percent, 5-year average)

Figure 5. Tajikistan: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	7	727.9
Breach of benchmark	No	Yes
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.